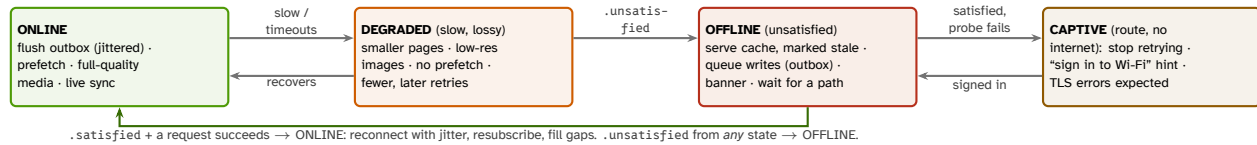


In one line: On a phone the network is a **variable**: it vanishes (tunnel), **degrades** (one bar, 20 % loss), **lies** (a captive portal answers every request with a login page), **costs money** (cellular, Low Data Mode) and **changes under you** (Wi-Fi → LTE). Give every request a **timeout**, a **retry rule**, an **idempotency story** and an **offline behaviour** — then prove it under a simulated bad network.

Connectivity as a state machine — the path says “a route exists”, only requests say “it works”



Orthogonal flags (any state): `path.isExpensive` (cellular, hotspot) → defer big downloads, ask before video · `path.isConstrained` (**Low Data Mode**) → no prefetch, no autoplay, lower quality · `Low Power Mode` → stop optional work. **Transitions are debounced** (1-2s) so a flapping path does not thrash the UI or trigger a reconnect storm.

Timeouts per layer

Layer	what bounds it
user action	your deadline: spinner 10s → cached data + Retry
URLSession idle	<code>timeoutIntervalForResource</code> (60s): max gap between bytes
whole transfer	<code>timeoutIntervalForResource</code> (default 7 days)
waiting for a path	<code>waitForConnectivity</code> : bound it yourself (cancel)
TCP connect	<code>NWProtocolTCP.Options.connectionTimeout</code>

Retries and idempotency

- Retry **transient** errors only (`timedOut`, `networkConnectionLost`, `cannotConnectToHost`, 502/503/504, 429 with `Retry-After`); never 4xx logic errors. Exponential backoff + full jitter + a cap.
- URLSession already retries **idempotent** requests when a reused connection drops (QA1941) — don't stack your own immediate retry on top.
- A timeout means **unknown**, not failed: POSTs carry an Idempotency-Key created once per user action and reused on every attempt.

Detecting connectivity

- `NWPathMonitor`: `status(.satisfied / .unsatisfied / .requiresConnection)`, `unsatisfiedReason`, `isExpensive`, `isConstrained` (iOS 13), `usesInterfaceType(.wifi)`. It sees *routes*, not the internet.
- Don't pre-flight**: attempt, handle the error; retry on *signals* (path satisfied, foreground), not only timers.
- Low Data Mode**: `allowsConstrainedNetworkAccess = false` makes the task fail with `URLError.networkUnavailableReason == .constrained` — retry with a lighter request (also `.expensive`, `.cellular`).
- Captive portals**: the OS probes a known URL (Apple: `captive.apple.com`) and shows the sign-in sheet; until then requests get redirects, HTML instead of JSON (decode errors on a “200”), or a certificate for the wrong host. Standard: RFC 8908/8910.

Example — path + session setup

```
let monitor = NWPathMonitor()
monitor.pathUpdateHandler = { path in
    Task { @MainActor in
        net.isOnline = path.status == .satisfied
        net.lowData = path.isConstrained // Low Data Mode
        net.metered = path.isExpensive // cellular, hotspot
        if net.isOnline { await outbox.flush() } // jittered inside
    }
}
monitor.start(queue: DispatchQueue(label: "net.path"))

let cfg = URLSessionConfiguration.default
cfg.waitsForConnectivity = true // wait, don't fail -1009
cfg.allowsConstrainedNetworkAccess = false // honour Low Data Mode
cfg.timeoutIntervalForResource = 20 // idle gap, not total
// delegate: URLSession(_:taskIsWaitingForConnectivity:) -> "Waiting..."
```

Related: `websockets-realtime` · `urlsession-networking` · `app-size-and-energy` · `api-design` (idempotency keys) · `sync-engines` (offline sync)

Coalescing, queuing, cancelling

- Single-flight**: one in-flight Task per cache key (an actor holding `[Key: Task<Value, Error>]`); five cells asking for the same avatar make one request.
- Cancel** what the user left: tie requests to the view's lifetime (`.task` modifier, stored Task cancelled in `deinit`) — bytes for a dismissed screen are waste.
- Prioritise**: `URLSessionTask.priority`; visible cells before prefetch; separate sessions for bulk vs interactive (`httpMaximumConnectionsPerHost`).
- Outbox** for writes: persist the intent first (optimistic UI), send in order with idempotency keys, remove on 2xx, surface permanent 4xx to the user. Background upload: a `.background` session (always waits for connectivity).
- Batch** small writes (analytics) — one radio wake, one request.

Testing bad networks

Network Link Conditioner	Mac: <i>Additional Tools for Xcode</i> ; iPhone: Settings → Developer. Profiles: 100% Loss, 3G, DSL, Edge, High Latency DNS, LTE, Very Bad Network, WiFi
Xcode Device Conditions	Devices window → Network Link / Thermal State (Xcode 11+, device on iOS 13+)
Android emulator	<code>-netdelay gsm edge umts lte min:max</code> , <code>-netspeed edge lte up:down(kbps)</code> ; <code>-netfast = none</code>
Toxiproxy	TCP proxy for tests/CI: latency, bandwidth, timeout, <code>reset_peer</code> , <code>slicer</code> , <code>limit_data</code> , <code>down toxics</code>

Plus Charles / Proxyman (throttle, breakpoints, fake 500s) and the real world (lift, train, hotel Wi-Fi). Script: offline launch, drop mid-upload, 30s latency, 5xx burst, captive, Low Data Mode — assert the *UI state*, not just “no crash”.

Traps

- ! “Check reachability, then request” — races, and a captive portal is “reachable”.
- ! Retry-on-timeout for a payment POST without an idempotency key → double charge.
- ! `timeoutIntervalForResource` is an *idle* timer: a slow trickle never trips it.
- ! Testing only in the simulator on office Wi-Fi: no radio, no loss, no NAT64.

Remember

“Attempt, don't ask; bound every wait; retry what is safe; queue what must survive.”